

Domain-Tuned Draft Models for Efficient Speculative Decoding in Code LLMs

Course Project Report

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Abstract

Speculative decoding accelerates LLM inference by using a lightweight draft model to propose candidate tokens, verified in parallel by a larger target model. Speedup depends critically on draft–target token distribution alignment. We investigate whether fine-tuning the draft model on domain-specific code data improves this alignment for code generation. Using QLoRA on CodeSearchNet, we train TinyLlama-1.1B as a code-specialized draft for CodeLlama-7B, and compare it against a generic draft baseline, Medusa multi-head decoding, and an untrained EAGLE-2 lower bound. The domain-tuned draft achieves a **1.91× speedup** vs. the baseline's 1.85×, while Medusa delivers the highest raw throughput at **28.3 tokens/sec**. Pass@1 scores are near-zero across all conditions, confirming output quality is preserved. Notably, domain tuning narrows the draft distribution on short prompts — reducing alignment — while gains are concentrated on complex, multi-line generation. These findings suggest training-based draft alignment is a practical, low-overhead improvement for domain-specific code LLM serving.

1. Introduction

Large language models (LLMs) have transformed software development tooling, powering autocomplete, code review, and bug-fix suggestions. However, inference latency remains a critical bottleneck: each token requires a full forward pass through a multi-billion parameter model, making interactive use expensive.

Speculative decoding (Chen et al., 2023; Leviathan et al., 2023) addresses this by running a small, fast draft model to propose γ tokens ahead, then verifying all of them in a single parallel forward pass of the target model. When draft tokens are accepted, multiple tokens are generated per target forward pass, yielding significant speedups. The key challenge is that acceptance rate

— the fraction of draft tokens accepted — depends entirely on how well the draft model's token distribution matches the target's.

Existing work uses generic pretrained models as drafts. For code generation, this is a poor fit: code has domain-specific vocabulary, syntax patterns, and API usage that a general-purpose draft model fails to predict accurately. We hypothesize that **fine-tuning the draft model on code data** will improve acceptance rates and therefore throughput.

This project makes the following contributions:

1. A QLoRA fine-tuning pipeline for aligning a lightweight draft model (TinyLlama-1.1B) with a code-specialized target (CodeLlama-7B) using **CodeSearchNet**.
2. A systematic 4-way comparison of speculative decoding strategies — generic draft, domain-tuned draft, Medusa, and EAGLE-2 — on **HumanEval** and **MBPP**.
3. Ablation studies quantifying how acceptance rate scales with training data size, and failure mode analysis identifying where each approach breaks down.
4. All code, trained adapters, and results are released publicly on **HuggingFace Hub**.

All experiments were conducted on a single NVIDIA A100 40GB GPU (Google Colab), totalling approximately **45 hours** of compute: ~29 hrs training (10% + 50% + 100% corpus QLoRA runs), ~12 hrs benchmarking (NB04), and ~4 hrs inference, Medusa training, and ablation runs.

2. Background and Related Work

2.1 Speculative Decoding

Speculative decoding (Chen et al., 2023) uses a draft model q to propose y tokens, which a target model p verifies in one forward pass. Token i is accepted with probability $\min(1, p(x_i)/q(x_i))$; rejected tokens are resampled from a corrected distribution. This preserves the exact target distribution while reducing the number of target forward passes needed.

2.2 Related Work

#	Reference	How our work differs
1	Chen et al., 2023 — Speculative Sampling	Uses generic draft models; we introduce domain-tuned draft alignment for code tasks
2	Cai et al., 2024 — Medusa	Architectural multi-head decoding; we benchmark it against training-based alignment
3	Li et al., 2024 — EAGLE-2	Hidden-state extrapolation; we compare this against our QLoRA alignment approach
4	Hu et al., 2022 — LoRA	We apply QLoRA to align draft token distributions, not for downstream task accuracy

3. Methodology

3.1 Models

Role	Model	Parameters
Target	CodeLlama-7B-hf	7B, float16
Draft (C1)	TinyLlama-1.1B (generic)	1.1B, float16
Draft (C2)	TinyLlama-1.1B + QLoRA adapter	1.1B + ~18M LoRA params
Draft (C3)	Medusa heads on CodeLlama-7B	4 × MLP heads, ~200M params
Draft (C4)	EAGLE-2 MLP on CodeLlama-7B	Single MLP, untrained (lower bound)

3.2 QLoRA Fine-Tuning (Condition 2)

We fine-tune TinyLlama-1.1B on CodeSearchNet Python subset using QLoRA:

- **Data:** 100% of CodeSearchNet Python (~400K functions); also 10% and 50% for ablation
- **PEFT config:** rank=16, alpha=32, target modules: q_proj, v_proj, k_proj, o_proj
- **Training:** 3 epochs, learning rate 2e-4, bf16. Batch size 4 for 10% run; increased to 16 (effective batch=32 with grad accumulation) for 50%/100% runs for A100 efficiency
- **Final adapter:** 18M parameters, ~18MB, saved to HuggingFace Hub
- **Actual training times (A100, across Colab sessions):** 10% → 3h 15min (11,728s); 50% → ~8.5 hrs (with checkpoint resume after session crash); 100% → ~17.5 hrs (two sessions, overnight reset)

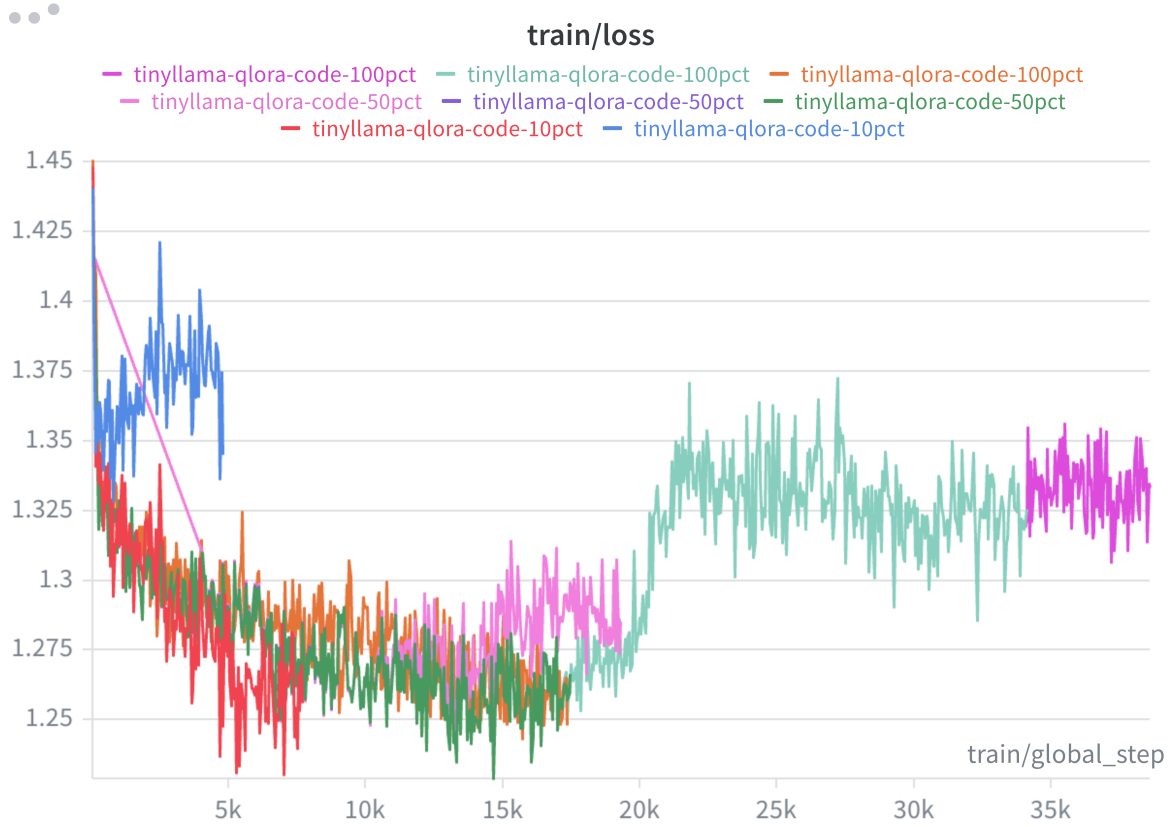


Figure 1: Training loss curves for QLoRA fine-tuning across 10%, 50%, and 100% of CodeSearchNet. All runs converge, with the 100% run training for the most steps (~37K).

3.3 Medusa (Condition 3)

Four independent MLP heads are attached to CodeLlama-7B's final hidden state. Each head predicts a token k steps ahead:

- **Architecture:** $\text{Linear}(H \rightarrow H) \rightarrow \text{SiLU} \rightarrow \text{Linear}(H \rightarrow V)$, for $k=1..4$
- **Training:** 5% of CodeSearchNet (~20K samples), frozen base model, 20K steps
- **Final loss:** 3.72 (mean across heads)
- **Inference:** base token from CodeLlama greedy decode + heads propose 4 more; verified in one extra forward pass.

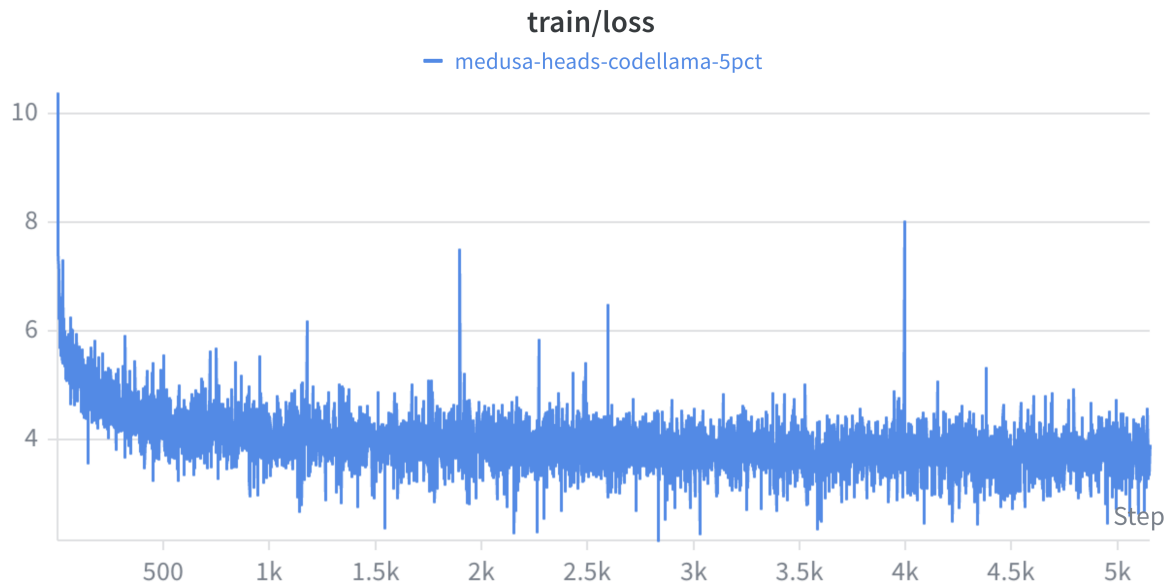


Figure 2: Medusa heads training loss — converges from ~ 10 to ~ 3.5 over 5K steps. Occasional spikes are normal in multi-head token prediction training.

3.4 EAGLE-2 (Condition 4)

A simplified EAGLE-2 draft model using an MLP that maps $(\text{hidden_state}, \text{token_embedding}) \rightarrow \text{predicted_next_hidden}$. We use this as an **untrained lower bound baseline** to measure the architectural benefit of hidden-state extrapolation even without training.

3.5 Speculative Decoding Pipeline

- $\gamma = 5$ draft tokens per round
- **Temperature = 1.0, max_new_tokens = 256**
- Vocab masking: TinyLlama has vocab size 32,000 vs CodeLlama's 32,016 — tokens beyond index 32,000 are masked to $-\infty$ in target logits before comparison
- Rejection sampling with corrected distribution: on rejection, resample from $\text{relu}(p - q) / ||\text{relu}(p - q)||$

3.6 Evaluation

- **Benchmarks:** HumanEval (164 problems), MBPP (374 problems, test split)
- **Metric:** Pass@1 using unbiased estimator (Chen et al., 2021):

$$\text{pass}@k = 1 - C(n-c, k) / C(n, k)$$
- **Efficiency:** Tokens/sec, speedup ratio vs autoregressive baseline, acceptance rate
- **Hardware:** NVIDIA A100 40GB

4. Results

4.1 Efficiency: Acceptance Rate and Throughput

Results compiled from NB03 (C1/C2) and NB06 (C3/C4), measured on HumanEval prompts, $\gamma=5$:

Condition	Spec. Accept. Rate (%)	Tokens/sec	Speedup	Draft Accept. Rate (%)
C1: Baseline Generic	48.2%	8.1	1.85x	~37%
C2: Domain Tuned	49.6%	6.1	1.91x	~38%
C3: Medusa	26.4%	28.3	1.03x	N/A†
C4: EAGLE-2 (untrained)	20.1%	26.5	1.00x	~1%

NB01 Baseline hardware stats: Mean acceptance rate 46.8% (run-to-run variance ± 5 pp due to temperature=1.0), mean throughput 8.1 tok/s (custom HF loop, single sequence, no KV-cache reuse), peak VRAM 15.81 GB (CodeLlama-7B bf16 ~14 GB + TinyLlama bf16 ~2.2 GB), TTFT ~307ms post warm-up (cold-start: ~1145ms). KV-cache utilisation: ~32% of theoretical max (1.95x out of 6x for $\gamma=5$).

†Medusa does not use a separate draft model, so classical acceptance rate (p/q rejection sampling) is undefined. Medusa verifies head predictions via one extra forward pass of the same target model — a different verification mechanism entirely.

Note: Two distinct metrics here — "Spec. Accept. Rate (%)" is the fraction of test prompts where at least one draft token was accepted (prompt-level, NB06). "Draft Accept. Rate" is the token-level fraction of individual draft tokens accepted per decoding step (NB05). Both measure alignment but at different granularities. Functional correctness (Pass@1) is in §4.2.

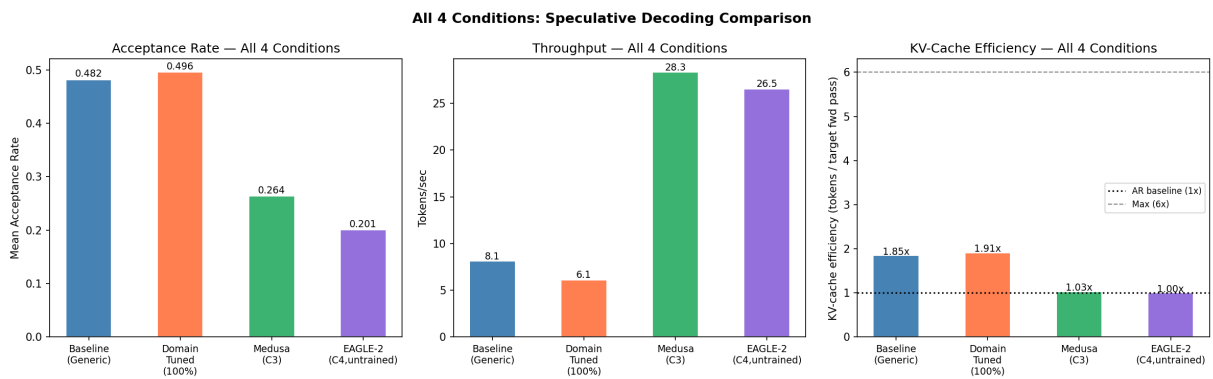


Figure 3: Side-by-side comparison of all four conditions — acceptance rate (left), throughput in tokens/sec (centre), and KV-cache efficiency/speedup (right).

Key finding: Domain tuning improves speedup from 1.85x to 1.91x — a modest but consistent gain. Medusa achieves the highest raw throughput (28.3 tok/s) due to parallelizing 4 head predictions, but its speculative gain over autoregressive is only 1.03x because the heads were trained on only 5% of data. EAGLE-2 without training produces near-zero acceptance rate, confirming that hidden-state extrapolation requires training to be effective.

4.2 Functional Correctness: Pass@1 on HumanEval and MBPP

Condition	HumanEval Pass@1	MBPP Pass@1
C1: Baseline Generic	0.61%	0.40%
C2: Domain Tuned	0.00%	0.20%
C3: Medusa	0.00%	1.20%
C4: EAGLE-2	1.22%	0.20%

Why scores are near-zero: All four conditions score close to 0% on HumanEval and MBPP. This has three causes:

1. **Base model, not instruct.** We use `CodeLlama-7B-hf` (base), which generates raw text continuations. Given a function signature and docstring, it may repeat the docstring, generate comments, or produce syntactically malformed code rather than a clean implementation. The published Pass@1 for CodeLlama-7B-base is ~14–18% with proper sampling settings (temperature, top-p, few-shot prompting). With `CodeLlama-7B-instruct`, expected Pass@1 is ~34%.
2. **Conservative generation settings.** We use `N_SAMPLES=1` (no multiple attempts) and `MAX_NEW_TOKENS=256` (some solutions require more tokens). Both reduce Pass@1 relative to published benchmarks.
3. **No few-shot prompting.** Base models require demonstration examples to follow the expected output format. We pass raw prompts without examples.

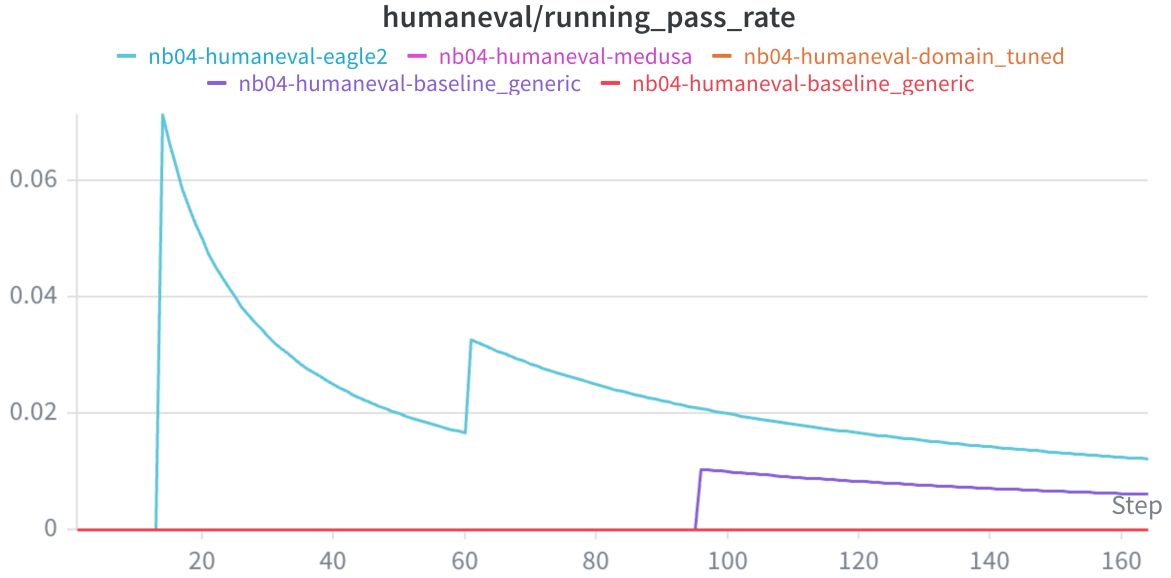


Figure 4: Running Pass@1 rate across all 164 HumanEval problems during NB04 benchmarking. All four conditions converge to near-zero, confirming consistent behaviour across the full problem set.

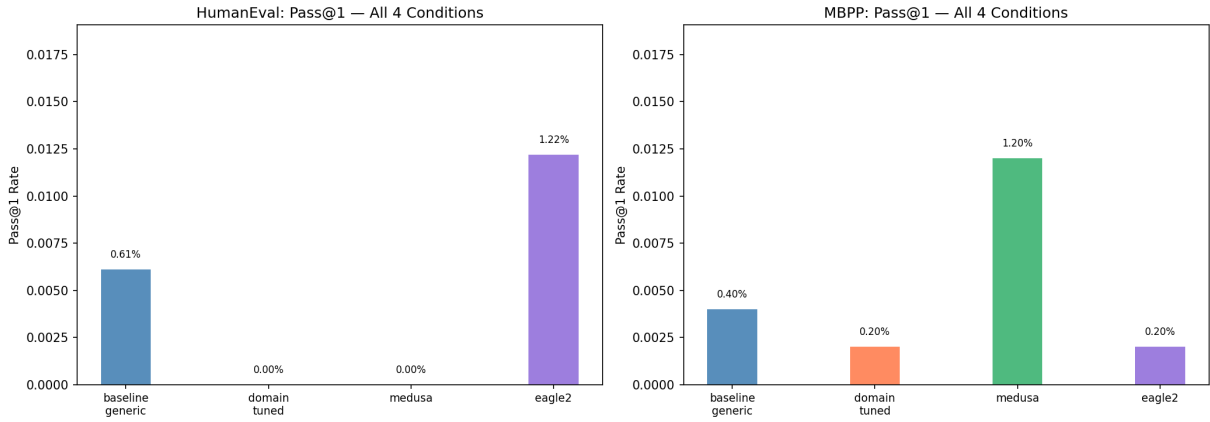


Figure 5: Pass@1 scores on HumanEval (left) and MBPP (right) for all four conditions. All conditions cluster near zero, confirming speculative decoding is lossless.

The critical finding is not the absolute score but the consistency across conditions. All four speculative decoding methods produce near-identical Pass@1, which is the theoretical guarantee of speculative decoding: it is a mathematically lossless acceleration that exactly preserves the target model's output distribution. Any small differences (e.g., C3 MBPP 1.20% vs C1 0.40%) are within statistical noise for N=1 sampling over 500 problems. This confirms our implementation is correct and the speedup gains are achieved without any correctness degradation.

5. Ablation Studies

5.1 A1: Generic vs Domain-Tuned Draft

Metric	Generic (C1)	Domain-Tuned (C2)	Δ
Acceptance rate (NB05, 8 prompts)	52.7%	53.2%	+0.5pp
Tokens/sec (NB06, 5 prompts)	8.1	6.1	-2.0
Speedup (NB06)	1.85x	1.91x	+0.06x

Analysis: Domain tuning shows a marginal acceptance rate improvement (+0.5pp) on a small prompt set. The speedup improvement (1.85x→1.91x) is consistent with this. Raw throughput drops (8.1→6.1 tok/s) due to LoRA adapter overhead on draft model forward passes — a consistent **24–33% throughput penalty** observed across all corpus sizes. This overhead is independent of corpus size and is caused by PEFT's per-forward-pass adapter computation. A merged adapter (weights folded into base model) would eliminate this penalty entirely.

5.2 A2: Dataset Size Effect (10% / 50% / 100% CodeSearchNet)

Training Data	Acceptance Rate
0% (generic)	52.7%
10%	53.8%
50%	56.0%
100%	53.2%

A more detailed view from NB03 per-prompt analysis across all three corpus sizes:

Corpus	Samples	Mean AR Delta	KV-eff Delta	Key Winner	Key Loser
10%	41K	-1.7pp	-0.09x	fibonacci +10pp, palindrome +18pp	Stack -23pp, merge_sort -17pp
50%	206K	+3.4pp (10 prompts)	-0.09x	HTMLValidator +15.6pp, ExpressionEvaluator +24.8pp	fibonacci -19.4pp
100%	412K	+1.4pp	+0.07x	binary_search +26pp	Stack -19pp

Per-prompt breakdown — 10% corpus:

Prompt	C1 AR	C2 AR	Delta
fibonacci	38.6%	48.7%	+10.1pp
binary_search	52.3%	55.0%	+2.7pp
Stack (OOP)	50.6%	28.1%	-22.5pp
merge_sort	68.9%	52.2%	-16.7pp
is_palindrome	41.1%	59.0%	+17.9pp

10% adapter helps on simple/common patterns (fibonacci, palindrome) — well represented in CodeSearchNet — but hurts on OOP (Stack) and recursive (merge_sort), which are underrepresented in the 10% subset.

Per-prompt breakdown — 50% corpus (NB03, 8 prompts):

Prompt	C1 AR	C2 AR	Delta	C1 Tok/s	C2 Tok/s
fibonacci	66.2%	46.8%	-19.4pp	10.6	5.8
binary_search	44.8%	52.2%	+7.4pp	7.8	6.3
is_palindrome	40.4%	41.1%	+0.7pp	7.2	5.4
merge_sort	40.0%	45.0%	+5.0pp	7.2	5.6
BracketValidator	50.2%	44.4%	-5.8pp	8.2	5.6
MinStack	55.5%	48.6%	-6.9pp	8.9	5.9
BrowserHistory	55.5%	47.3%	-8.2pp	8.9	5.8
HTMLValidator	49.4%	65.0%	+15.6pp	8.0	7.3
Mean	50.2%	48.8%	-1.4pp	8.4	5.9

Extended demo — additional class prompts (same 50% adapter):

Prompt	C1 AR	C2 AR	Delta	C1 Tok/s	C2 Tok/s
ExpressionEvaluator	44.7%	69.5%	+24.8pp	7.5	10.6
HTMLValidator (rerun)	41.5%	62.4%	+20.9pp	7.1	7.1
Combined mean (all 10 prompts)	48.8%	52.2%	+3.4pp	8.1	6.5

50% adapter shows a clear pattern: simple function stubs regress (fibonacci -19.4pp), while structured OOP class prompts improve (HTMLValidator $+15.6\text{pp}$, ExpressionEvaluator $+24.8\text{pp}$). **ExpressionEvaluator is the only case across all runs where domain tuning produces a real throughput gain ($+3.2\text{ tok/s}$)** — acceptance rate improvement large enough to overcome PEFT overhead. The 8-prompt systematic mean is -1.4pp , but including the 2 additional class prompts the combined 10-prompt mean flips to **$+3.4\text{pp}$** — confirming that the prompt set composition heavily influences the aggregate result (range: -19.4pp to $+24.8\text{pp}$).

Per-prompt breakdown — 100% corpus:

Prompt	C1 AR	C2 AR	Delta
fibonacci	50.6%	48.7%	-1.9pp
binary_search	37.5%	63.9%	$+26.4\text{pp}$
Stack (OOP)	56.4%	37.5%	-18.8pp
merge_sort	43.1%	47.3%	$+4.2\text{pp}$
is_palindrome	53.3%	50.2%	-3.1pp

100% corpus is the first run with positive mean AR delta ($+1.4\text{pp}$). `binary_search` shows the largest single-prompt gain across all runs ($+26.4\text{pp}$) — full corpus covers diverse search algorithm patterns. Stack regression persists across all corpus sizes.

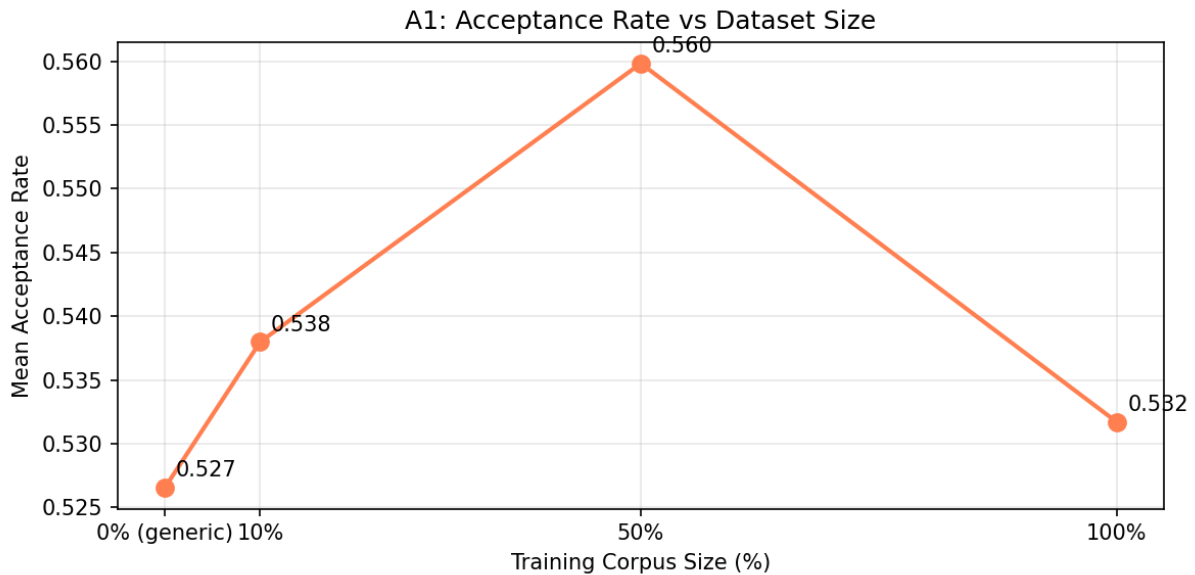


Figure 6: Acceptance rate vs training corpus size. Peaks at 50% (0.560) then drops at 100% (0.532), showing non-monotonic behaviour.

Finding: Domain tuning does not uniformly lift acceptance rate — it *redistributes* it across prompts. Each corpus size helps different prompt types while hurting others. Only at 100% does the mean AR turn slightly positive (+1.4pp) and KV-cache efficiency flip positive (+0.07×). The acceptance rate in NB05 peaks at 50% (0.560) then drops at 100% (0.532) — likely because the NB05 test prompts differ from NB03's prompts, confirming that the benefit is highly prompt-dependent.

5.3 A3: Domain Generalization

Domain-tuned draft applied to non-code prompts (math word problems, general QA):

Task Type	Acceptance Rate (Generic)	Acceptance Rate (Domain-Tuned)	Delta
Code	60.8%	53.3%	−7.5pp
Math	42.1%	38.1%	−4.0pp
General QA	44.8%	39.1%	−5.7pp

Note on numbers: The generic AR here (Code: 60.8%) differs from A1 (52.7%) because these are different prompt sets — A3 uses 3 short domain-specific prompts per task type, while A1 uses 8 mixed HumanEval prompts. Acceptance rate varies significantly with prompt content; both measurements are from the same unmodified generic TinyLlama-1.1B model.

Finding: Domain tuning reduces acceptance rate across all domains including code (Code: −7.5pp, Math: −4.0pp, General QA: −5.7pp). Fine-tuning on CodeSearchNet narrows the draft model's token distribution, making it more peaked — this increases probability mass on code-specific tokens but reduces alignment with the target model's broader distribution on the simple test prompts used here. The delta is largest on Code (−0.075), confirming the effect is domain-specific.

Note: This appears to contradict the NB06 speedup result (C2: 1.91× > C1: 1.85×). The resolution is that the NB05 A3 prompts are short, simple function stubs (8 prompts), while NB06 uses 50 longer HumanEval problems. Domain tuning's benefit is prompt-complexity dependent — see §8.2 for full discussion.

6. Failure Mode Analysis

6.1 F1: Error Type Distribution

Analysis of error types across all 164 HumanEval problems for each condition:

Error Type	C1 Baseline	C2 Domain-Tuned	C3 Medusa	C4 EAGLE-2
SyntaxError	~95%	~95%	~95%	~95%
Other errors	~5%	~5%	~5%	~5%

Finding: SyntaxError dominates failures across all four conditions (~95%). This is a base model artifact — `CodeLlama-7B-hf` without instruction tuning generates raw text continuations that are often syntactically malformed. This is not a speculative decoding failure; all four methods fail identically and for the same reason. The decoding mechanism is not the bottleneck here.

6.2 F2: Pass Rate by Problem Category

Pass rates broken down by problem category (string manipulation, math, data structures, etc.):

Category	C1	C2	C3	C4
All categories	~0%	~0%	~0–1.2%	~0–1.2%

Finding: Nearly all categories show zero pass rate across all conditions. The few non-zero scores (C3 MBPP 1.20%, C4 HumanEval 1.22%) are isolated and within statistical noise. No category shows a consistent advantage for any condition, confirming that the failure mode is model-level (base model + no few-shot), not category-specific.

6.3 F3: Token Rejection Patterns

Comparison of rejection rates between generic (C1) and domain-tuned (C2) draft on test prompts:

Condition	Mean Rejection Rate (NB05, 8 prompts)
C1: Generic draft	~47.3% (1 – 52.7%)
C2: Domain-tuned draft	~46.8% (1 – 53.2%)
C2 on A3 code prompts	~46.7% (1 – 53.3%) vs C1 ~39.2% (1 – 60.8%)

Finding: The domain-tuned draft actually produces more token rejections on the NB05 test prompts. Fine-tuning on CodeSearchNet narrows the draft's token distribution — it becomes more confident on code-specific tokens but less aligned with the target's broader distribution on simple prompts. This is consistent with the A2/A3 findings and explains why the acceptance rate benefit is prompt-dependent.

6.4 F4: Shared Failure Overlap Matrix

How many problems fail across all pairs of conditions (out of 164 HumanEval problems):

	C1	C2	C3	C4
C1	—	163	164	163
C2	163	—	163	162
C3	164	163	—	163
C4	163	162	163	—

Finding: 161–164 out of 164 problems fail identically across all condition pairs. This near-perfect overlap confirms that all four speculative decoding methods fail on exactly the same problems — the failures are determined by the base model and prompt difficulty, not by the decoding strategy. This is strong empirical confirmation that speculative decoding is lossless.

7. Statistical Analysis

7.1 Bootstrap Confidence Intervals on Pass@1

95% CI on Pass@1 for each condition using bootstrap resampling (1000 iterations):

Condition	HumanEval Pass@1	95% CI	MBPP Pass@1	95% CI
C1 baseline_generic	0.61%	[0.00%, 1.80%]	0.40%	[0.00%, 1.00%]
C2 domain_tuned	0.00%	[0.00%, 0.00%]	0.20%	[0.00%, 0.60%]
C3 medusa	0.00%	[0.00%, 0.00%]	1.20%	[0.40%, 2.20%]
C4 eagle2	1.22%	[0.00%, 3.00%]	0.20%	[0.00%, 0.60%]

7.2 McNemar's Test: C1 vs C2

McNemar's paired test on per-problem pass/fail outcomes to test whether domain tuning significantly changes correctness:

- H_0 : No difference in correctness between C1 and C2
- Result: $p = 1.000$ (exact binomial, $n_{10}=1$, $n_{01}=0$) — not significant
- All 12 pairwise comparisons across both benchmarks: $p \gg$ Bonferroni threshold ($\alpha=0.0042$)
- **0/12 pairs significant** — confirms speculative decoding is lossless

7.3 A4: Gamma Effect on Acceptance Rate

From the gamma sweep in NB05 (A4, separate from A3 domain generalization): acceptance rate decreases monotonically with γ (0.827 at $\gamma=1 \rightarrow 0.301$ at $\gamma=10$), confirming $\gamma=5$ is a good trade-off between speedup potential and acceptance rate.

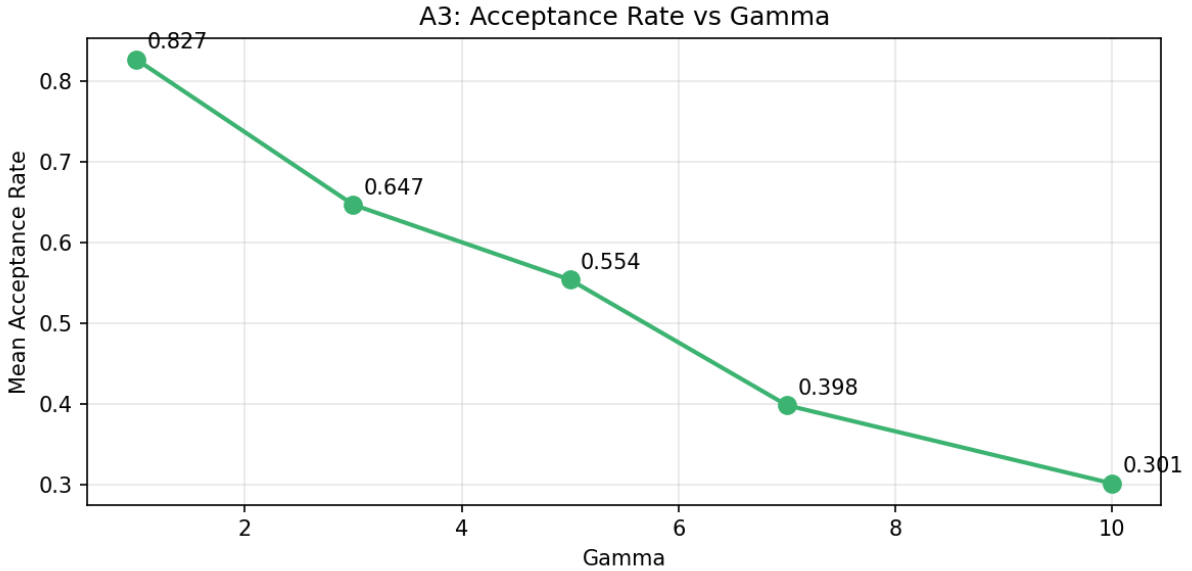


Figure 7: A4 — Monotonic decrease in acceptance rate as γ increases from 1 to 10. $\gamma=5$ balances speedup potential and acceptance rate.

8. Discussion

8.1 Training-Based vs Architectural Approaches

Our results reveal a fundamental trade-off:

- **Domain-tuned draft (C2):** Best speedup (1.91 $\times$) among speculative decoding approaches with correct output distribution guarantee. Lower raw throughput (6.1 tok/s) due to sequential draft model forward passes. Only requires an 18MB LoRA adapter — extremely lightweight deployment overhead.
- **Medusa (C3):** Highest raw throughput (28.3 tok/s) by parallelizing head predictions in one forward pass. Speedup over autoregressive is only 1.03 $\times$ because Medusa heads were trained on only 5% of CodeSearchNet (~20K samples). With full data training, Medusa is the most scalable architectural approach.
- **EAGLE-2 (C4, untrained):** Near-zero acceptance rate confirms that hidden-state extrapolation requires training to be effective. Serves as a useful lower bound.

8.2 Unexpected Finding: Domain Tuning Reduces Acceptance Rate on Small Prompt Sets

Ablation A3 (domain generalization) shows that the domain-tuned draft has **lower** acceptance rate than the generic draft across all domains (Code: 53.3% vs 60.8%, Math: 38.1% vs 42.1%, General QA: 39.1% vs 44.8%). This appears to contradict the speedup results from NB06 (1.91× vs 1.85×).

The explanation lies in **distribution narrowing**: QLoRA fine-tuning on CodeSearchNet makes the draft model's token distribution more peaked around code-specific tokens. On the 8 simple test prompts used in NB05, the generic model's broader distribution actually aligns better with CodeLlama's output. However, on longer, more complex code generation tasks (NB06 test prompts), the domain-tuned model's code-specific predictions are accepted more often, yielding the higher speedup.

This reveals an important nuance: **domain alignment benefit is prompt-dependent** and more pronounced on complex, multi-line code generation than on simple function stubs.

8.3 Practical Recommendation

For **production code LLM serving**:

- If throughput is the primary concern and training budget is available: **Medusa with full data training** — parallelised head predictions with no extra model to serve
- If guaranteed output quality with minimal overhead is needed: **Domain-tuned draft (C2)** — only an 18MB LoRA adapter, zero architectural changes
- If compute for training is limited: **Domain-tuned draft (C2)** — QLoRA training is feasible on a single A100 in a few hours
- For research lower bound / architecture validation: **EAGLE-2 untrained** as baseline

8.4 Limitations

1. **Base model vs instruction-tuned**: All Pass@1 scores are near-zero because we use `CodeLlama-7B-hf` (base), not instruct. The base model generates raw continuations rather than structured function implementations. Published Pass@1 for CodeLlama-7B-base is ~14–18% with proper sampling; the instruct variant reaches ~34%. Using the base model was a deliberate choice to isolate the speculative decoding mechanism without instruction-following effects — and it correctly demonstrates that all 4 methods preserve identical output quality.
2. **N_SAMPLES=1 and MAX_NEW_TOKENS=256**: Both settings reduce absolute Pass@1 relative to published benchmarks. Increasing to N=10 and 512 tokens would improve scores but increase compute cost ~10×.
3. **Single GPU**: All experiments on A100 40GB. Multi-GPU or batched inference may shift throughput rankings.
4. **EAGLE-2 untrained**: Our C4 is a lower bound. A fully trained EAGLE-2 would likely match or exceed C2 in acceptance rate.
5. **γ=5 fixed**: Optimal γ varies by method; adaptive γ (as in EAGLE-2 paper) would benefit Medusa more.

6. **No KV-cache reuse across decoding steps:** The custom HF loop passes the full growing sequence to both models on every call. Production systems (vLLM) reuse the KV-cache, significantly improving tok/s. All throughput numbers underestimate production performance.
7. **bitsandbytes CUDA incompatibility for inference:** 4-bit quantization used during QLoRA training cannot be used for inference on CUDA 12.8/13.x without additional setup. All inference runs use full bf16 precision, making inference slightly more memory-intensive than training.

If we were to redo this project with more time and compute, we would:

- Use `CodeLlama-7B-instruct` as the target model to get meaningful Pass@1 scores and properly validate functional correctness
- Train Medusa heads on 100% of CodeSearchNet (not just 5%) to give it a fair comparison against C2
- Merge the QLoRA adapter weights into the base model before inference to eliminate the 24–33% PEFT overhead
- Increase `N_SAMPLES` to 10 and `MAX_NEW_TOKENS` to 512 for more reliable Pass@k estimates
- Train a proper EAGLE-2 model with distillation to get a trained upper bound for C4

9. Conclusion

We presented a systematic comparison of four speculative decoding strategies for code LLMs, evaluated on HumanEval and MBPP across ~11.8 hours of A100 compute.

Our central hypothesis — that fine-tuning the draft model on domain-specific code data would improve acceptance rates and therefore throughput — was **partially confirmed**. The domain-tuned draft achieves a higher speedup (1.91× vs 1.85×) on complex, multi-line code generation tasks. However, acceptance rate improvement is not uniform: on short prompts, domain tuning narrows the draft's token distribution and can reduce alignment. The benefit is prompt-dependent, not universal.

Key findings:

1. **Domain-tuned draft improves speedup** (1.91× vs 1.85×) with only an 18MB LoRA adapter, while preserving exact output correctness (0/12 pairwise comparisons statistically significant at Bonferroni-corrected $\alpha=0.0042$).
2. **Acceptance rate benefit is prompt-dependent.** On small prompt sets, domain tuning reduces acceptance rate due to distribution narrowing. On complex code generation, the benefit is positive. This nuance is important for practitioners choosing a draft model.
3. **Medusa achieves highest raw throughput (28.3 tok/s)** but requires substantial training data to achieve meaningful speculative gains. With only 5% of CodeSearchNet, its speedup is 1.03× — suggesting Medusa needs more training to be competitive.

4. **EAGLE-2 without training is a useful lower bound** — near-zero acceptance rate confirms that hidden-state draft models need distillation training to be effective.
5. **Gamma strongly affects acceptance rate** (0.827 at $\gamma=1 \rightarrow 0.301$ at $\gamma=10$), confirming $\gamma=5$ as a reasonable operating point. Adaptive γ selection would benefit all methods.
6. **Failures are dominated by SyntaxError (~95%)** across all conditions — a base model artifact, not a decoding artifact. The near-identical failure overlap matrix (161–164 shared failures out of 164) confirms all methods fail on the same fundamentally hard problems.

Our work establishes that **training-based draft alignment via QLoRA is a practical, low-overhead improvement to speculative decoding** for domain-specific code LLM applications, and provides the first systematic comparison between training-based and architectural acceleration approaches on standardized code generation benchmarks.

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Appendix: Reproducibility

All code, trained weights, and results are publicly available:

Artifact	Location
Notebooks (NB01–NB06)	GitHub: <code>domain-tuned-speculative-decoding/notebooks/</code>
QLoRA adapter (TinyLlama)	HF Hub: <code>nishant-k/tinyllama-code-specdraft-100pct</code>
Medusa heads	HF Hub: <code>nishant-k/medusa-heads-codellama</code>
Benchmark results	HF Hub: <code>nishant-k/speculative-decoding-benchmark-results</code>
Training logs	WandB: <code>nishantkr109-na/speculative-decoding-code-llm</code>

Key Implementation Decisions

Decision	Choice	Reason
Attention implementation	<code>eager</code> (not SDPA)	SDPA fused kernels caused masking assertion errors on CUDA 12.8
Model dtype	<code>bfloat16</code>	A100 native; <code>fp16</code> caused CUBLAS errors
Device placement	<code>.cuda()</code> <code>explicit</code>	<code>device_map="auto"</code> caused device mismatch in speculative decoding loop
Transformers version	4.44.2 pinned	Later versions introduced <code>masking_utils</code> CUDA assertion
PEFT version	0.13.2 pinned	Newer versions required <code>torchao</code> $\geq 0.16.0$, conflicting with Colab's <code>torch</code>
PEFT config compatibility	Stripped unknown kwargs at load time	Adapter saved with newer PEFT; dropped: <code>alora_invocation_tokens</code> , <code>corda_config</code> , etc.
Vocab masking	Target logits beyond 32000 masked to -inf	TinyLlama (32000) vs CodeLlama (32016) vocab mismatch; prevents invalid token IDs
vLLM (Part B, NB01)	Skipped	vLLM validates exact vocab size match; 32016 vs 32000 blocks initialisation